

30/07/26

Activity

① A CCTV image contains salt-and-pepper noise due to transmission errors.

Technique used :- Median Filter

Reason :- The median filter effectively removes salt-and-pepper noise by replacing each noisy pixel with the median value, while preserving image edges.

② A Satellite image is corrupted by Gaussian noise from sensors.

Technique used : Gaussian Filter.

Reason : Smooths Gaussian noise effectively and reduces random sensor noise.

③ A medical x-ray image has random noise but fine edges must be preserved.

Technique used : Bilateral Filter.

Reason : Removes noise while preserving important edges.

④ A Scanned document has black and white dots scattered randomly.

Technique used : Median filter

Reason : Eliminates black and white dot noise without affecting text.

⑤ A low-light image shows heavy grain without clear boundaries.

Technique used : Adaptive Median filter

Reason : Reduces heavy grain and preserves image details better.

③ A foggy road image has very low contrast

Technique used : Histogram Equalization

Reason : Enhances contrast and improves visibility.

④ An under water image appears dull and washed out.

Technique used : Contrast stretching

Reason : Increases image contrast and improves brightness.

⑤ A chest x-ray shows poor visibility of internal structures.

Technique used : CLAHE [Contrast Limited Adaptive Histogram Equalization]

Reason : Enhances local contrast and reveals internal structures clearly.

⑥ A night-time traffic image lacks intensity variation.

Technique used : Histogram Equalization

Reason : Expands local contrast and reveals internal structures.

⑩ A gray scale image uses only a narrow intensity range.

Technique used : Contrast stretching

Reason : Expands the narrow intensity range to improve overall image contrast.